

model (discussed for a binary fit in Section 6.2). Because there are three possible choices, the appropriate data model (applied at each SOA) is no longer the binomial distribution, but rather the multinomial distribution, which can provide an exact likelihood of obtaining any particular combination of probabilities that divide N choices into three bins when the actual probabilities of selecting each bin are known (or rather, for fitting purposes, predicted).²²

11 Dual-Presentation SJ Data

Several authors have investigated the use of a dual-presentation SJ task in which two bimodal stimuli are presented (one after another) and compared, for example by reporting which one was (most) synchronous (Allan & Kristofferson, 1974; Powers, Hillock, & Wallace, 2009; Roseboom, Nishida, Fujisaki, & Arnold, 2011). This is a form of what would, in classical signal detection theory, be described as a two-alternative forced choice (specifically the two-interval forced choice variant). However, that designation is ambiguous (about whether there are two presentations or two response categories) and has been applied to cases where either or both of the possible qualifying conditions are met, which is probably why the dual-presentation SJ task has ended up being given a variety of names (e.g., temporal 2AFC; forced-choice successiveness discrimination; 2IFC SJ, where the classic SJ is referred to as 2AFC SJ in the same paper). I will label it the 2xSJ.

The simplest form of the 2xSJ would have a synchronous standard on every trial along with a non-synchronous test pair. Based on the kind of observer models discussed in this chapter, the resulting psychometric function (plotting the probability of judging the standard more synchronous than the test against the test's SOA) is U-shaped and centred over the PSS. This approach represents a reasonable way to derive estimates of inverse precision (i.e., $\sigma_{\Delta t}$) but a fairly poor way to estimate the PSS, because having a synchronous standard on every trial provides feedback about objective synchrony. A simple solution is to also include a range of standards as well as a range of tests, in a roving standard design.

The observer model can be fitted to data even when both standard and test are non-zero, as described in detail by Yarrow et al. (2016; see also García-Pérez & Peli, 2014). To present all of the data, it is necessary to plot a function for each standard SOA (using several standard plots, or a single 3D plot), which is somewhat cumbersome, but not a major obstacle to using the task. A simple

22 <MultinomialLikelihood g>.